Amar Prakash Pandey

Senior Data Engineer

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Summary

Senior Data Engineer with **8+ years** of experience **designing scalable, high-performance data architectures**, **ELT/ETL pipelines**, and **large-scale batch processing systems**. Proficient in handling **petabyte-scale datasets** with **Apache Spark, Databricks, Airflow, and dbt**, backed by strong **SQL, Python, and Scala**. Skilled in building **cloud-native data solutions** on **AWS and GCP**, with a focus on **performance optimisation, cost efficiency, and data quality**.

Contributor to **Apache Airflow**; past maintainer at **CRI-O and DuckDuckGo**. Writes about **Spark internals and performance tuning** at amarpandey.me/blog

Experience

Citi — Pune, India

Senior Data Engineer — June 2025 – Present

- Cut Spark job resource usage by **up to 90%** by root-causing configuration issues in failing, over-provisioned production jobs; built Spark log and Autosys JIL analyzers that speed up diagnosis team-wide.
- Built **var-compare** from scratch, a daily validation tool that reconciles Market Risk numbers before reports are published.
- Led the **Maven → Gradle** migration of the trading-reports mono-repo: live in 2 weeks with **zero incidents**, **~40% faster builds**, and vulnerable dependencies removed.
- Initiated test automation on a zero-test ~20K-LOC codebase, reaching **60% coverage**; it became a team-wide mandate (80% minimum in CI). Codified team conventions as **Claude Code skills** so AI-generated code stays consistent across the team.

Sahaj Software — Pune, India

Senior Data Engineer — March 2021 – June 2025

Developed and maintained end-to-end batch pipelines to process audience and frame-level data for out-of-home (OOH) advertising, enabling accurate and timely delivery of insights to campaign optimisation and analytics teams. Owned data quality checks and pipeline performance end-to-end for effective ad targeting and impact measurement.

- **Designed and optimised ETL pipelines** with **Apache Spark (PySpark) on Databricks**, processing terabytes of data and executing large-scale joins to enrich advertisement analytics.
- **Enhanced Spark job performance by 50%** through advanced shuffle partition tuning and resource optimisation, significantly reducing processing time.
- **Implemented and managed** workflow orchestration with **Apache Airflow**, ensuring efficient sequencing, automated retries, and SLA compliance for batch processing.
- **Cut S3 storage costs by 60%** by implementing efficient data lifecycle policies and optimising file formats for cost-effective cloud storage.
- Automated deployment of **Airflow DAGs**, **Apache Spark/PySpark jobs**, and **microservices** using **GitLab CI/CD**, built with **Cursor AI** to deliver the setup in a fraction of the estimated time with broader test and edge-case coverage, eliminating manual errors and ensuring seamless promotion across dev and production environments.

Tata Consultancy Services — Chennai, India

System Engineer — December 2017 – February 2021

Developed a **pricing platform** that streamlined and automated pricing updates for thousands of components. My contributions focused on **optimizing workflows, improving system reliability, and enhancing deployment efficiency**.

- **Developed APIs** for Cummins' pricing platform using **Java, Spring Boot, and MySQL**, enabling real-time pricing updates for thousands of components.
- **Automated pricing workflows**, reducing update time from **one day to near real-time**.
- **Migrated Oracle trigger and function logic** to a service layer for **better logging and maintainability**.
- Automated deployment using CI/CD pipelines with Git, **Jenkins, and GitLab**, improving efficiency and reliability.

Skills

Programming Languages

Java, Python, Scala, SQL

Big Data Technologies

Apache Spark, PySpark, Databricks, Apache Hadoop

ETL Tools & Orchestration

Apache Airflow, dbt (Data Build Tool)

Cloud Technologies

AWS S3 EMR, ECS, ECR, Lambda, RDS, Google Cloud Platform BigQuery

Databases

PostgreSQL, Redshift, DynamoDB, MongoDB, Oracle

DevOps

Git, Docker, Terraform, Jenkins, GitLab CI/CD

Projects

spot-optimizer January 2025

Built and published an **open-source Python library on PyPI** that replaces in-house logic for **AWS Spot Instance selection** and recommends the most stable, cost-effective instances based on cores, memory, region, and **EMR version compatibility**.

geohashviz.com July 2024

GeohashViz is a web-based visualization tool that helps users explore **geohashes** interactively. It allows users to encode locations into geohashes, visualize different precision levels, and understand spatial hierarchies

Finger Detection and Tracking April 2017

A computer vision project (**400+ GitHub stars**) that detects and tracks fingers in real time using **OpenCV and Python**, enabling gesture-based interactions from a live camera feed.

Awards

Senior Data Engineer March 2024

Recognized for exceptional contributions at **Sahaj.ai**, receiving multiple commendations for assisting projects, building useful tools, and consistently going above and beyond.

System Engineer June 2019

Cummins

Awarded Star Performer of the Year at **TCS** for outstanding performance on the **Cummins** project.

Software Developer August 2017

Google Summer of Code

Completed Google Summer of Code (2017), contributing to the **Teammates project** at the **National University of Singapore**.